# Abhinav Ramalingam

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## ABOUT

**Passionate AI Engineer focused on applying data science and machine learning to real-world business problems, with hands-on experience analysing ambiguous requirements and translating complex, structured/unstructured data into actionable insights. Skilled at evaluating and choosing the right methodological approach for a given problem, and at communicating technical findings. Practical exposure to multi-agent frameworks, cloud platforms such as GCP, AWS, and Azure, and generative AI models including GPT, Gemini, and Claude**

## EDUCATION

**Uppsala University**, Uppsala, Sweden 09/2024 – 06/2026
M.Sc., Computer Science
**Relevant courses:** Large Language Models, Natural Computation Methods, Deep Learning, Statistical and Probabilistic ML, Data Mining, Data Security and Privacy

**SSN College of Engineering**, Chennai, TN, India 06/2020 – 05/2024
B.E., Computer Science & Engineering CGPA | 8.66
**Relevant courses:** Artificial Intelligence, Machine Learning, Foundations of Data Science, Software Development and Object Oriented Design, Linear Algebra and Calculus

## EXPERIENCE

**DeepLore Technologies** *Summer Intern 2025* Chennai, India (06/2025 – 08/2025)

- Built MCP and API tooling to collect compliance evidence for GRC workflows from cloud platforms (Azure, Entra IAM, AWS, GCP), version control (GitHub, GitLab) and vulnerability scanners (Nessus, Secpod), contributing to the open-source framework, SuperAgentX.
- Collaborated with peers to design and test custom handlers for GRC-focused tooling, bridging the gap between raw governance-risk-compliance data and actionable, structured outputs for downstream workflows.
- Built multi-stage agentic workflows using SuperAgentX for use cases (agentic software development, regulatory document assistant), integrating memory, reasoning, and prompt templates to enable downstream development.

**Purple Cube Technologies** *SDE Intern 2024* Chennai, India (02/2024 – 05/2024)

- Built a CRM application covering Lead Management, Customer Information, Quotation Management, and Project Tracking modules.
- Designed relational data models and normalised schemas for leads, customers, quotations, and project records, ensuring referential integrity and queryable structured outputs.
- Implemented data integration logic and report generation pipelines, reducing reliance on manual, unstructured data handling across business workflows.

## PROJECTS

**Comparative Analysis of Methods for Transforming EHR Data to Canonical Knowledge Graph Elements** *Master Thesis, KTH Royal Institute of Technology*

- Identified graph data structures from literature to select the optimal representation for EHR data.
- Researched and identified appropriate structural profiling approaches in terms of accuracy, memory and runtime performance.
- Implemented different approaches for entity-event classification and multi-entity table decomposition
- Benchmarked and compared traditional automated approaches and LLM-based approaches for extracting canonical knowledge graph elements; evaluated on real-world de-identified clinical text data (MIMIC-IV).

**LLM-Powered Career Assistant** *Large Language Models, M.Sc., Computer Science*

- Built a career assistant with three action paths: General Q&A, Career Coach, and Cover Letter Generation.
- Implemented a TF-IDF RAG system for context-aware cover letter generation using uploaded resumes and templates.
- Integrated Google Calendar OAuth to automatically schedule shortlisted job opportunities from live job board searches.
- Designed a modular FastAPI back-end with Pydantic validation, specialised routers, and a RAG maintenance layer.
- Evaluated cover letter quality using JD similarity (TF-IDF cosine), latency benchmarking, and structural metrics.

**QURIO: AI-Powered Interview Intelligence Platform** *National Start-up Buildathon, 2026*

- Designed an end-to-end LLM-powered interview intelligence system that transforms raw transcripts into evidence-based insights.
- Includes a AI-driven, modifiable question-generation system with in-form response analysis and follow-up question generation.
- Built a synthesis engine that breaks responses into traceable atomic units and evaluates them against user-supplied hypotheses.
- Built with Typescript backend; Pitched the project idea to a panel involving a VC firm member and Uppsala University AI society

**TinyStoriesLM: A Replicated Study** *Transformers, Language Models*

- Replicated the TinyStories study by training decoder-only Transformer language models from scratch at three scales ($\sim$0.1M, 0.8M, and 4.7M non-embedding parameters) on 20K synthetic stories.
- Implemented a byte-level BPE tokenizer and self-supervised next-token prediction pipeline with AdamW, learning-rate decay, and gradient clipping; evaluated models using validation perplexity and LLM-based quality assessment.
- Analysed the effect of model scale on language generation, observing monotonically decreasing perplexity and a qualitative improvement in grammar, consistency, and plot coherence at the largest model.

**Bayesian Inference in Diffusion Tensor Imaging** *Probabilistic ML, M.Sc., Computer Science*

- Implemented and compared four Bayesian inference methods (Metropolis-Hastings, Importance Sampling, Variational Inference, and Laplace Approximation); estimated posterior distributions from real-world MRI data.
- Developed Variational Inference using mean-field approximation with Gamma and Wishart variational posteriors.
- Implemented Laplace Approximation with Cholesky reparametrisation to enforce positive-definiteness of the diffusion tensor.
- Evaluated scalability of all methods for full-brain analysis of $\sim 10^5$ voxels, concluding Laplace Approximation offers the best balance of computational efficiency and approximation quality.

**GPU Optimisation of Sparse Matrix-Vector Multiplication** *Accelerator-Based Programming, M.Sc., Computer Science*

- Benchmarked Sparse Matrix-Vector Multiplication for a Conjugate Gradient solver using CRS and SELL-C-$\sigma$ formats on NVIDIA GPU via CUDA.
- Implemented cuSPARSE-based SpMVM as a performance baseline, and custom SELL-C-$\sigma$ kernel to demonstrate tailored memory access patterns for FEM-structured sparsity.
- Analysed format conversion and host-device transfer overhead, proposing amortisation strategies and GPU-side conversion to mitigate SELL-C-$\sigma$ setup costs.

## SKILLS

- **Development**: *Programming* (Python, JavaScript, C/C++, Bash), *Frameworks* (FastAPI, Django, Flask, ReactJS, NextJS), *Cloud & DevOps* (Docker, Kubernetes, AWS, Azure, GCP), *Version Control* (Git, GitHub, GitLab), Testing and CI/CD (PyTest, Github Actions), AI-assisted programming (Claude Code, Github Copilot, Lovable, Gemini AI Studio)
- **Databases**: ER Diagrams, SQL (Postgres, SQLite), NoSQL (Neo4j, MongoDB)
- **AI/ML:** *ML Frameworks* (Numpy, Scipy, Scikit-Learn, Pandas, Matplotlib, PyTorch, Tensorflow, Keras), *LLMs* (GPT, Gemini, Claude, Mistral, Deepseek, LLama, Hugging Face), *Agentic AI* (RAG, Prompt Engineering, MCP), AI/ML Orchestration (Airflow, MLflow)